4b AIDS CSBS CSE(AIML) CE CSE(Cyber) ECE IT — ES

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Semester: III | **B. Tech. Artificial Intelligence and Data Science/B. Tech. in Computer Science & Business Systems/B. Tech. Computer Science and Engineering (Artificial Intelligence and Machine Learning)/ B. Tech. Computer Engineering/B. Tech. Computer Science and Engineering (Cyber Security)/ B. Tech. in Electronics and Computer Engineering/B. Tech. Information Technology** | **39524304**

2413ADU3T1/
2413CBU3T1/
2413CAU3T1/
2413CEU3T1/
2413CCU3T1/
2413ECU3T1/
2413ITU3T1

Embedded Systems

Date: 09/06/2026 **Time: 02:00 PM TO 04:00 PM** **Max Marks: 60**

Q.1	**Attempt all Questions**	**20 Marks**	**CO**	**BT**
a)	Explain the working of D-type Flip-Flop with circuit and truth table.	4	CO1	BT2
b)	Describe MOV, MOVX and MOVC instructions in 8051 Microcontroller.	4	CO3	BT2
c)	Write a program to ADD two 8-bit numbers stored in external RAM.	4	CO4	BT4
d)	Discuss DC motor interfacing with 8051 Microcontroller with diagram.	4	CO5	BT2
e)	Elaborate CPSR register in ARM Microcontroller.	4	CO6	BT3
Q.2	**Solve any Four**	**20 Marks**	**CO**	**BT**
a)	Evaluate the conversion of decimal number (625) to its Hexadecimal equivalent number.	5	CO1	BT3
b)	Compare Microprocessor and Microcontroller.	5	CO2	BT4
c)	Explain the internal memory organization of 8051 microcontroller.	5	CO3	BT2
d)	Illustrate the different addressing modes in 8051 Microcontroller.	5	CO4	BT4
e)	Discuss LCD interfacing with 8051 Microcontroller with diagram.	5	CO5	BT2
f)	Describe various operating modes in ARM Microcontroller.	5	CO6	BT2

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Q.3	Solve any Two	20 Marks	CO	BT
a)	Illustrate the Multiplexer and De-Multiplexer with circuit diagram.	10	CO1	BT4
b)	Formulate and describe a case study of washing machine.	10	CO2	BT5
c)	Write an assembly language program to Unpack the packed BCD Number.	10	CO4	BT4